Supplementary Material

Lowest Error, Best Decision? Accuracy Equivalence and Lead-Time-Dependent Rankings of Remaining Useful Life Models for Turbofan Maintenance

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S0 Resampling and Statistical Procedure

Partition and seeds. Each subset was split into five engine-grouped folds by an unshuffled GroupKFold, which is deterministic; seeds 42 to 46 initialised the five outer fold models (random forest, gradient boosting and CNN) and seeds 142 + 10k + c the inner pair models, so they do not affect the partition. Test-set accuracy is reported over the five fold models (mean ± SD, Table S1); paired accuracy tests use the fold-averaged prediction for each test engine.

Nested calibration. For outer fold k, models fitted on the other four folds predict the run-to-failure trajectories of fold k (outer predictions). For each calibration fold $c \neq k$, models fitted on the three folds other than k and c predict fold c (inner predictions); 10 pair fits and 5 outer fits per model and subset. Preprocessing (regime clustering, scaling, variance screen) was refitted in every fit. For engine i and lead L, the required margin r(i, L) is the smallest m for which the first cycle with predicted RUL $\leq$ L + m has true RUL $\geq$ L; because the alarm time is monotone in m, the zero-miss margin for fold k is the largest r(i, L) over its calibration engines and the 2% margin is the $(\lfloor 0.02n \rfloor + 1)$-th largest.

Bootstrap. In each of B = 2,000 replicates, engines were drawn with replacement within each outer fold; every fold's margin was recomputed from the resampled calibration engines' inner required margins (models were not refitted), held-out engines were re-evaluated at the new margins, and the paired difference $\Delta*$ in mean unused life over engines alerted in time by both models was recomputed. Intervals are the 2.5th and 97.5th percentiles of $\Delta*$; $p = \min\{1, 2 \min[P(\Delta* \leq 0), P(\Delta* \geq 0)]\}$, so its resolution is 0.001 and p = 0 is reported as $p < 0.001$.

Multiplicity. The confirmatory family is the 15-cycle lead with the zero-miss target: RF − GB, GB − LR and CNN − GB in each subset, Holm-adjusted within the subset (three contrasts). Table S8 also gives Holm adjustment across all 12 contrasts. All other leads, the 2% target, the repeated partitions and the cap-150 analysis are exploratory; repeated partitions reuse the same engines and are not independent replications.

S1 Full Accuracy Results

Table S1. Official-test accuracy (capped labels) over the five fold models (mean ± SD), of the fold-averaged prediction, the number of folds (of five) in which each tabular model ranked first among the four tabular models (ties split), and the nested out-of-fold (OOF) RMSE over all training trajectories used by the decision lens.

Subset	Model	MAE	RMSE	NASA	Fold-avg. MAE / RMSE / NASA	RMSE wins	NASA wins	Nested OOF RMSE
FD001	LR	16.25 ± 0.25	20.60 ± 0.30	1105 ± 83	16.23 / 20.56 / 1096	0	3	20.69
	Ridge	16.20 ± 0.25	20.56 ± 0.29	1106 ± 82	16.17 / 20.52 / 1098	0	0	20.70
	RF	13.13 ± 0.26	19.06 ± 0.52	1350 ± 213	12.63 / 18.66 / 1257	0	0	18.81
	GB	12.91 ± 0.56	18.44 ± 0.70	1133 ± 190	12.72 / 18.23 / 1070	5	2	18.44
	CNN	12.83 ± 0.36	16.70 ± 0.45	777 ± 188	11.99 / 15.43 / 602	–	–	15.49
FD002	LR	15.01 ± 0.12	18.82 ± 0.11	1765 ± 46	14.99 / 18.80 / 1757	0	0	20.48
	Ridge	15.00 ± 0.12	18.81 ± 0.11	1758 ± 45	14.97 / 18.78 / 1751	0	0	20.48
	RF	12.10 ± 0.18	16.06 ± 0.17	1486 ± 28	12.05 / 15.87 / 1431	5	5	18.80
	GB	12.35 ± 0.14	16.51 ± 0.18	1639 ± 87	12.27 / 16.39 / 1591	0	0	18.72

Subset	Model	MAE	RMSE	NASA	Fold-avg. MAE / RMSE / NASA	RMSE wins	NASA wins	Nested OOF RMSE
	CNN	15.28 ± 0.96	20.11 ± 1.09	3067 ± 701	13.82 / 17.11 / 1612	–	–	20.14
FD003	LR	16.04 ± 0.24	20.27 ± 0.31	1493 ± 185	16.00 / 20.20 / 1471	0	2	18.71
	Ridge	15.99 ± 0.22	20.24 ± 0.31	1495 ± 181	15.96 / 20.18 / 1474	0	2	18.71
	RF	13.95 ± 0.24	19.90 ± 0.57	1777 ± 299	13.50 / 19.38 / 1505	4	1	17.11
	GB	14.17 ± 0.43	19.90 ± 0.61	1889 ± 211	13.97 / 19.66 / 1807	1	0	16.62
	CNN	11.19 ± 0.85	15.55 ± 1.11	653 ± 179	10.31 / 14.10 / 443	–	–	13.65
FD004	LR	17.17 ± 0.14	21.09 ± 0.20	2360 ± 139	17.15 / 21.06 / 2340	0	5	19.52
	Ridge	17.17 ± 0.14	21.09 ± 0.20	2361 ± 139	17.15 / 21.06 / 2342	0	0	19.52
	RF	13.86 ± 0.14	19.07 ± 0.15	3327 ± 462	13.74 / 18.87 / 3109	5	0	17.31
	GB	14.10 ± 0.12	19.29 ± 0.16	3270 ± 500	14.04 / 19.16 / 3149	0	0	17.23
	CNN	13.67 ± 0.77	19.30 ± 1.07	3529 ± 1485	12.24 / 16.88 / 1814	–	–	17.54

S2 Diagnostic of the CNN near Failure

Without a margin the CNN missed 97 to 100% of engines at a 15-cycle lead although it was the most accurate model out of fold on FD001 and FD003. Table S2 and Fig. S1 show why: its nested out-of-fold predictions have a floor near failure. Median predictions at failure were 16 to 25 cycles with narrow interquartile ranges on FD001, FD003 and FD004, so the predicted RUL rarely or never reaches 15 cycles while the engine still has 15 cycles left.

Implementation checks. (i) Window alignment: each 30-cycle window ends at the current cycle; padding (repeating the first observation) affects only the first 29 cycles of an engine, far from failure. (ii) Rescaling: the network was trained on RUL/125 and its output multiplied by 125; predictions for healthy engines (true RUL ≥ 150, capped label 125) averaged 118 to 123 cycles. (iii) Bias pattern: a misalignment or rescaling error would shift or scale predictions throughout life; here the bias falls from about +16 to +23 cycles at failure to +5 to +12 at a true RUL of 30 to 50 (Table S2), with a small IQR at failure. (iv) In-sample check: refitting fold 0 of FD001 and predicting the training engines themselves gave the same floor (Table S3), so it is not an out-of-sample effect. (v) Ablation: removing the 0.5 dropout before the dense layer removed the floor (median 2.8 cycles at failure) with slightly better held-out fold RMSE, whereas training for 300 instead of 150 epochs raised it. The floor is therefore a property of this dropout configuration (as in Li et al.), not an implementation error, and the CNN's decision results hold for this configuration.

Table S2. Distribution of nested out-of-fold predicted RUL at selected true (uncapped) RUL values over all training engines. Bias: mean predicted minus true RUL. Min pred.: lowest prediction anywhere in the trajectories. Healthy mean: mean prediction at true RUL ≥ 150.

Subset	Model	Pred. at RUL 0: median [IQR]	Pred. at RUL 15: median [IQR]	Share ≤ 15 at RUL 15	Pred. at RUL 30: median [IQR]	Bias at 30 / 50	Min pred.	Healthy mean
FD001	LR	−10.2 [−28.8, 3.4]	22.3 [12.9, 33.7]	0.33	48.3 [37.4, 59.7]	+18.4 / +19.8	−94.7	115.2
	RF	1.7 [1.2, 2.6]	17.2 [13.5, 22.9]	0.33	34.2 [27.8, 48.8]	+9.9 / +14.2	0.5	115.5
	GB	4.2 [2.2, 5.9]	17.8 [13.8, 22.4]	0.35	34.6 [27.3, 44.8]	+9.0 / +14.0	−6.1	115.8
	CNN	18.6 [17.6, 20.0]	24.6 [22.9, 26.3]	0.00	35.7 [32.1, 40.6]	+6.7 / +4.8	15.1	119.7
FD002	LR	−9.2 [−16.0, −3.3]	23.3 [14.6, 31.2]	0.26	47.2 [38.5, 56.6]	+17.9 / +20.9	−89.6	114.8
	RF	2.3 [1.7, 3.4]	17.7 [13.8, 22.3]	0.34	36.2 [28.1, 46.4]	+9.6 / +16.2	0.6	115.2
	GB	4.2 [2.7, 5.7]	17.7 [13.0, 22.1]	0.38	35.4 [27.8, 46.5]	+9.3 / +16.5	−9.4	115.4
	CNN	24.8 [16.1, 28.1]	28.7 [23.6, 32.0]	0.00	38.2 [32.8, 44.0]	+9.7 / +12.1	11.8	117.6
FD003	LR	−16.3 [−28.8, 6.1]	23.1 [15.8, 33.6]	0.23	47.5 [37.4, 56.5]	+15.9 / +19.4	−69.2	119.3
	RF	1.9 [1.3, 2.9]	16.7 [14.6, 21.1]	0.30	34.8 [28.7, 44.8]	+7.2 / +13.1	0.6	118.7
	GB	3.7 [1.2, 5.5]	17.5 [12.1, 22.3]	0.39	32.5 [28.0, 44.6]	+6.1 / +13.9	−5.4	119.0
	CNN	15.8 [15.2, 17.0]	21.8 [20.2, 23.9]	0.03	34.7 [31.1, 40.0]	+5.8 / +6.4	9.0	123.3
FD004	LR	−6.6 [−22.2, 5.7]	26.6 [15.8, 37.2]	0.22	47.8 [38.0, 60.2]	+18.7 / +20.5	−62.7	118.6
	RF	2.7 [1.8, 4.0]	18.7 [15.2, 24.4]	0.24	36.6 [30.3, 46.8]	+9.2 / +14.9	1.0	118.5

Subset	Model	Pred. at RUL 0: median [IQR]	Pred. at RUL 15: median [IQR]	Share ≤ 15 at RUL 15	Pred. at RUL 30: median [IQR]	Bias at 30 / 50	Min pred.	Healthy mean
	GB	4.2 [1.6, 7.0]	18.9 [14.3, 24.5]	0.28	35.6 [28.5, 45.9]	+8.3 / +14.7	−3.3	118.4
	CNN	20.4 [18.2, 21.6]	25.3 [23.0, 27.9]	0.00	37.2 [31.2, 43.4]	+8.2 / +11.1	14.5	120.9

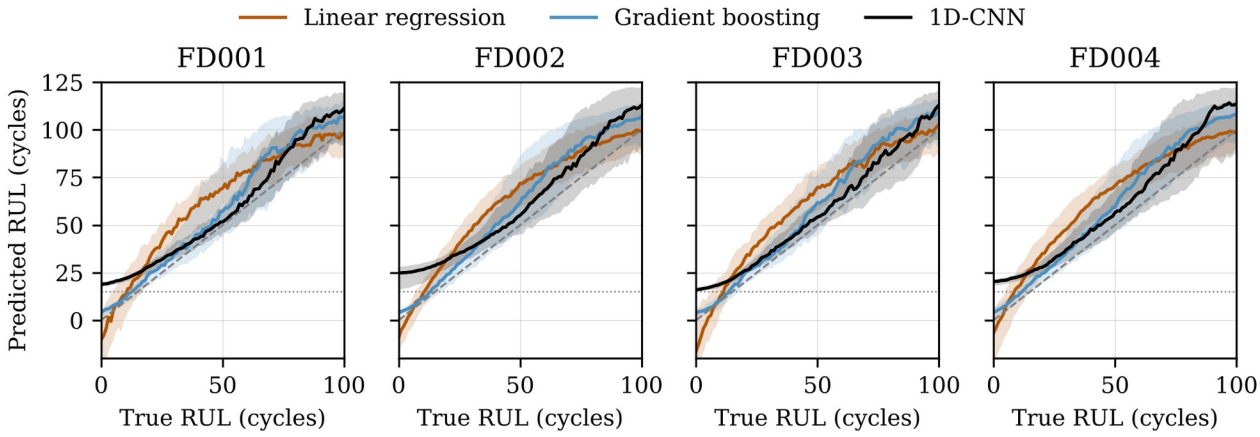


Fig. S1. Median (line) and interquartile range (band) of nested out-of-fold predicted RUL against true RUL for linear regression, gradient boosting and the CNN. Dashed: perfect prediction; dotted: 15-cycle alarm level at m = 0.

Table S3. In-sample check and ablation for the CNN on FD001 (one fit on folds 1 to 4, seed 42). RMSE against capped labels over all cycles of the engines shown.

Run (FD001, folds 1–4 fitted)	Engines	Median pred. at RUL 0	at RUL 15	at RUL 30	Min pred.	RMSE
As in paper (dropout 0.5, 150 epochs)	training (in-sample)	18.3	23.4	34.6	17.4	8.72
	held-out fold 0	18.2	21.7	32.8	17.5	17.40
No dropout, 150 epochs	training (in-sample)	1.4	15.0	31.9	−6.7	7.27
	held-out fold 0	2.8	14.7	29.1	−3.4	16.11
Dropout 0.5, 300 epochs	training (in-sample)	27.7	31.1	38.6	27.2	8.92
	held-out fold 0	27.7	30.3	38.0	27.2	19.08

S3 Why Linear Regression Wins at Long Lead Times

Table S4 decomposes the margins. Near failure (L = 10 or 15) the ensembles' out-of-fold errors at true RUL = L are tight (SD about 4 to 8 cycles) compared with linear regression (13 to 16), so they need smaller margins. At L = 30 their spread is similar to linear regression's, but the upper tail is heavier: the zero-miss margin, set by the hardest one or two calibration engines, exceeds the 95th-percentile requirement by 16 to 42 cycles against 13 to 18 for linear regression, and more calibration engines are outlying. On FD001, FD002 and FD004 the resulting alarm level L + m (91 to 109 cycles) lies below, and close to, the ensembles' predictions for healthy engines (115 to 119 cycles; Table S2), so alarms fire as soon as predictions leave that plateau, at a median true RUL of 94 to 112 cycles against 74 to 81 for linear regression. Unused life is computed as the true RUL at the alarm minus L, that is, assuming removal L cycles after the alarm. At the 95th-percentile margin (last column) the linear advantage shrinks on FD002 and reverses on FD004, but not on FD001, so the extreme calibration engines drive much, though not all, of the linear advantage at L = 30.

Table S4. Margin decomposition by lead time L. Zero-miss margins: range over the five outer folds. P95 req.: 95th percentile of the calibration engines' required margins (mean over folds). Tail gap: zero-miss margin minus P95 requirement (mean over folds). Outlying engines: calibration engines with required margin above median + 3 IQR (mean per fold). Error SD: SD of out-of-fold prediction error at true RUL = L. Median alarm RUL: median true RUL at the alarm for timely-alerted held-out engines. Unused at P95 margin: unused life if each fold used its P95 requirement as margin.

L	Subset	Model	Zero-miss margins (folds)	P95 req. (mean)	Tail gap	Outlying engines	Error SD at RUL = L	Median alarm RUL	Unused, zero-miss	Unused at P95 margin
10	FD001	LR	27–32	22.6	6.0	0.2	15.5	25	18.4	14.6
		RF	14–16	10.0	4.4	0.0	4.5	20	12.2	8.4
		GB	11–16	9.8	3.8	1.0	4.5	20	12.6	9.5
		CNN	15–16	13.4	2.2	0.0	1.9	17	7.4	4.7
	FD002	LR	27–31	21.2	8.4	0.0	12.6	26	17.8	12.7
		RF	17–20	12.2	5.8	0.0	5.3	24	15.7	10.3
		GB	17–20	11.2	7.6	1.8	5.2	25	17.0	10.4
		CNN	23–27	20.8	4.8	0.0	5.5	26	20.5	15.0
	FD003	LR	35–44	27.6	10.2	1.6	15.7	33	25.7	17.4
		RF	12–17	8.6	5.0	0.2	4.2	20	12.2	8.2
		GB	11–16	9.4	4.0	1.8	5.2	20	12.3	9.3
		CNN	16–43	13.8	9.4	0.8	4.2	25	21.2	8.1
	FD004	LR	32–41	27.2	8.8	0.0	15.8	30	22.6	16.8
		RF	16–35	12.6	15.4	0.8	6.8	34	26.6	13.0
		GB	19–27	13.4	9.4	0.6	6.3	30	23.2	14.2
		CNN	27–31	19.4	9.2	1.0	3.4	35	29.4	16.3
15	FD001	LR	31–34	26.4	6.2	0.0	14.1	31	20.2	16.4
		RF	19–23	16.6	5.0	0.0	6.9	32	18.9	14.5
		GB	19–27	14.8	6.6	0.4	6.6	35	20.8	15.8
		CNN	13–17	12.6	2.8	0.0	2.8	23	9.2	6.1
	FD002	LR	42–49	27.0	19.4	0.2	13.5	44	32.8	15.2
		RF	24–31	16.6	9.8	0.6	7.1	34	22.9	15.1
		GB	28–32	15.4	14.2	3.4	7.4	37	26.0	14.4
		CNN	25–44	19.8	13.8	0.4	5.7	44	37.1	15.6
	FD003	LR	40–52	29.2	16.0	0.0	15.4	44	32.6	18.3
		RF	17–25	13.0	8.2	1.2	5.7	33	19.7	12.4
		GB	14–22	13.0	5.0	0.0	6.9	34	18.5	13.6
		CNN	16–59	14.6	12.6	0.8	5.1	32	27.0	11.3
	FD004	LR	40–50	33.2	11.2	0.0	15.0	40	30.3	20.4
		RF	27–38	16.8	14.0	1.0	7.4	41	28.9	16.3
		GB	21–31	16.0	10.6	0.2	7.5	37	26.6	16.9
		CNN	24–73	19.4	16.4	0.6	4.2	43	42.1	18.2
30	FD001	LR	49–54	33.6	17.4	0.0	14.5	74	60.9	26.8
		RF	64–70	46.0	21.4	3.8	17.9	98	78.3	47.4
		GB	61–73	42.0	26.0	7.0	17.6	98	79.1	39.5
		CNN	19–26	16.0	6.8	0.0	6.6	55	28.5	16.5
	FD002	LR	55–60	39.4	17.8	0.0	15.0	81	67.1	29.7
		RF	72–76	42.0	31.2	6.2	17.1	112	93.9	38.1
		GB	69–74	38.6	33.0	7.4	17.0	102	85.5	33.0
		CNN	41–76	25.0	27.4	2.2	11.1	95	79.7	32.7
	FD003	LR	44–53	34.2	13.0	0.0	15.0	68	42.7	23.8
		RF	38–54	31.4	15.6	2.6	14.0	68	50.9	30.5
		GB	32–58	30.2	18.6	4.8	13.3	66	45.4	28.6
		CNN	19–65	14.6	16.0	0.4	8.1	54	34.4	14.1
	FD004	LR	53–59	44.4	12.8	0.0	15.9	78	68.2	38.6

L	Subset	Model	Zero-miss margins (folds)	P95 req. (mean)	Tail gap	Outlying engines	Error SD at RUL = L	Median alarm RUL	Unused, zero-miss	Unused at P95 margin
		RF	70–79	31.6	41.8	2.8	14.8	111	97.9	31.5
		GB	67–77	35.4	35.6	3.6	15.0	94	79.4	31.7
		CNN	38–58	23.2	21.4	1.0	9.8	81	62.9	33.6

S4 Out-of-Fold Miss-Rate Curves

For each model, the margin was chosen on the pooled nested out-of-fold trajectories as the smallest that leaves at most a fraction q of engines missed, and the unused life of timely-alerted engines was recorded (in-sample with respect to the margin, so descriptive). At L = 15 the tabular models differed by at most 6.3 cycles at q = 5% but by up to 15.5 cycles at q = 0, and the step from 5% to zero misses accounted for 21 to 55% of their zero-miss unused life: models separate in the tail of the error distribution.

Table S5. Unused life on the pooled nested out-of-fold trajectories at 5% and 0% allowed misses, spread between the four tabular models, and the share of zero-miss unused life due to the step from 5% to zero misses (range over tabular models).

L	Subset	LR 5% / 0%	Ridge 5% / 0%	RF 5% / 0%	GB 5% / 0%	CNN 5% / 0%	Tabular spread 5% / 0%	Step share, tabular
10	FD001	14.5 / 18.4	14.3 / 19.1	8.6 / 12.8	9.6 / 10.1	6.4 / 10.5	5.9 / 9.0	5–33%
	FD002	12.4 / 18.2	12.3 / 18.2	9.5 / 14.7	10.1 / 16.2	16.5 / 22.9	2.8 / 3.5	32–38%
	FD003	17.2 / 25.7	17.1 / 25.6	8.3 / 12.0	9.4 / 13.0	7.3 / 41.5	8.9 / 13.8	28–33%
	FD004	16.0 / 22.0	16.6 / 22.0	13.3 / 28.4	13.7 / 27.2	14.6 / 24.8	3.4 / 6.5	24–53%
15	FD001	15.7 / 21.3	16.5 / 21.2	15.4 / 19.4	15.3 / 22.4	8.8 / 13.3	1.1 / 3.0	21–32%
	FD002	15.6 / 33.3	15.5 / 34.4	15.5 / 20.7	14.8 / 23.2	17.0 / 30.8	0.7 / 13.7	25–55%
	FD003	17.8 / 35.2	17.8 / 35.2	11.5 / 20.6	12.8 / 19.7	9.1 / 46.4	6.3 / 15.5	35–49%
	FD004	19.4 / 30.4	19.4 / 30.3	16.4 / 28.7	16.3 / 27.7	15.3 / 28.0	3.1 / 2.7	36–43%
30	FD001	27.8 / 62.3	26.8 / 64.6	48.5 / 68.1	39.5 / 76.7	18.9 / 24.4	21.7 / 14.3	29–59%
	FD002	30.5 / 67.1	30.5 / 67.0	38.9 / 95.9	34.4 / 86.3	34.3 / 84.5	8.4 / 28.9	55–60%
	FD003	23.4 / 43.8	23.3 / 43.9	31.6 / 44.7	28.8 / 47.9	14.9 / 64.2	8.3 / 4.2	29–47%
	FD004	40.0 / 70.2	40.1 / 70.1	31.3 / 92.0	31.0 / 73.8	34.1 / 43.3	9.1 / 21.9	43–66%

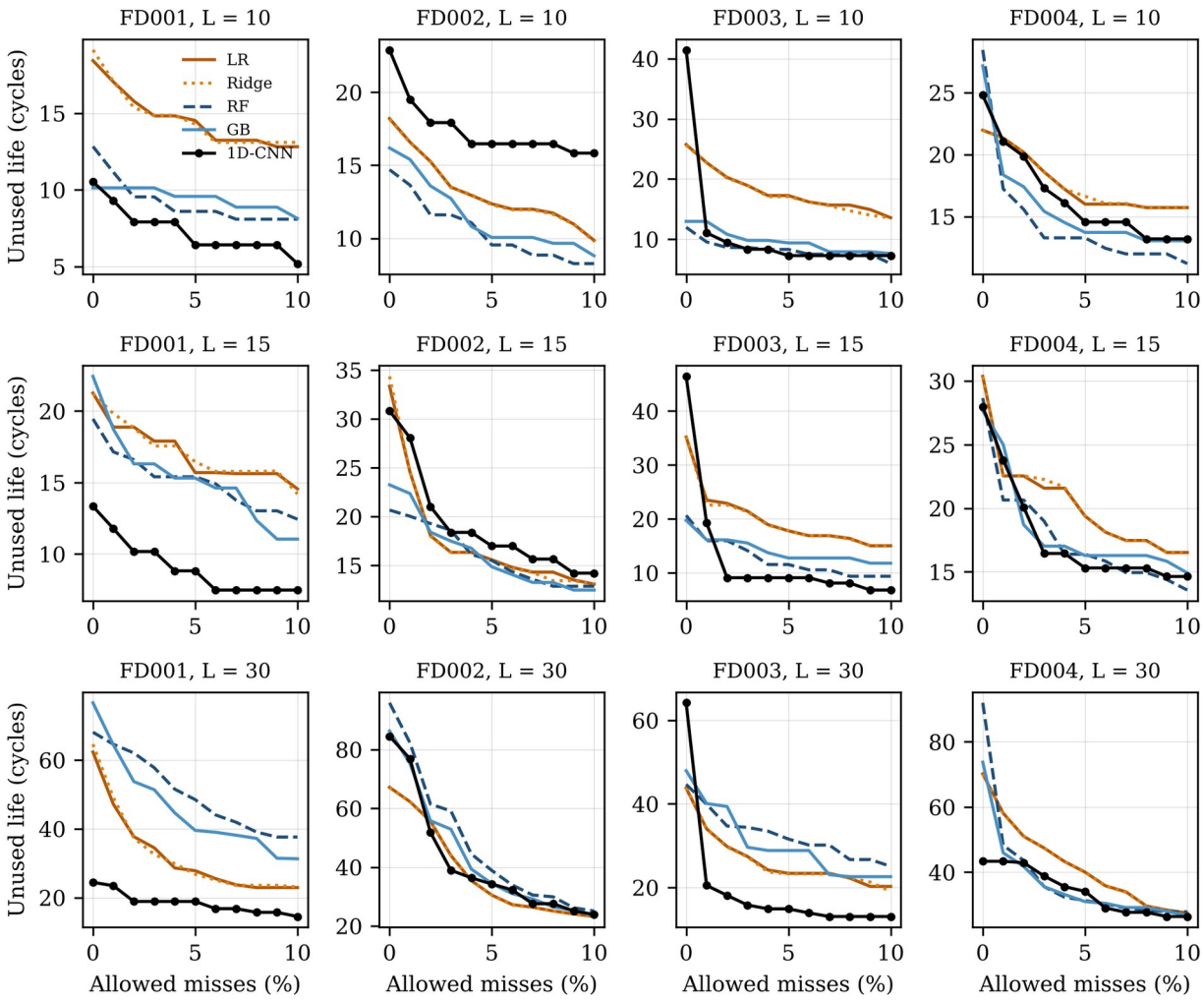


Fig. S2. Unused life against allowed miss rate on the pooled nested out-of-fold trajectories, by subset (columns) and lead time (rows).

S5 Significant Contrasts across Partitions

Table S6. Counts of significant paired differences in unused life (95% interval excluding zero) by partition. Linear vs ensemble: GB − LR and RF − LR, two targets and four subsets per lead (16 cells per partition, 48 in total). RF vs GB: eight cells per lead and partition (24 in total). The partitions reuse the same engines, so these counts are not independent replications.

Contrast type	Lead L	Favoured	Nested	P1	P2	Total (of 48 or 24 cells)
Linear vs ensemble	10	ensemble	11	10	10	31 of 48
	15	ensemble	4	2	3	9 of 48
	30	linear	7	2	5	14 of 48
	any	opposite direction	–	–	–	0
RF vs GB	10	RF	1	0	2	3 of 24
	15	RF	1	1	0	2 of 24
	30	GB	1	2	2	5 of 24
	any	opposite direction	–	–	–	0

S6 Sensitivity to a Penalty per Missed Engine

Table S7. Nested decision lens over all 24 cells (3 leads × 2 targets × 4 subsets): comparison of the cost per engine (unused life of timely-alerted engines plus c per missed engine, divided by all engines) with the paired unused-life comparison. At c = 400, 16 of the 17 significant tabular ensemble advantages at 10 and 15 cycles lost significance, while all 7 significant linear advantages at 30 cycles remained.

Pairs	c (cycles per miss)	Significant, kept	Lost significance	Reversed	Newly significant
Tabular (RF − GB, GB − LR, RF − LR)	25	25	0	0	0
	100	21	4	0	0
	400	9	16	0	0
CNN vs tabular	25	28	1	0	2
	100	17	12	0	4
	400	9	20	0	4

S7 Primary Contrasts and Multiplicity

Table S8. Confirmatory family: 15-cycle lead, zero-miss target. Negative Δ favours the first model. With $p < 0.001$ (no replicate crossing zero), adjusted values are bounds.

Subset	Contrast	Δ unused [95% CI]	n	p	Holm within subset	Holm across 12
FD001	RF − GB	−1.9 [−4.6, 1.9]	98	0.585	1.000	1.000
	GB − LR	0.6 [−3.8, 2.6]	98	0.961	1.000	1.000
	CNN − GB	−11.9 [−15.5, −7.5]	96	< 0.001	< 0.003	< 0.012
FD002	RF − GB	−3.1 [−5.0, −1.3]	260	0.010	0.030	0.110
	GB − LR	−6.7 [−7.8, 1.8]	259	0.223	0.223	1.000
	CNN − GB	11.0 [−1.4, 13.6]	259	0.096	0.192	0.768
FD003	RF − GB	1.2 [−2.8, 3.3]	98	0.674	1.000	1.000
	GB − LR	−14.1 [−17.6, −2.7]	98	0.011	0.033	0.110
	CNN − GB	8.6 [−5.6, 11.6]	97	0.720	1.000	1.000
FD004	RF − GB	2.5 [−4.7, 5.1]	246	0.527	1.000	1.000
	GB − LR	−3.8 [−9.3, 3.5]	246	0.606	1.000	1.000
	CNN − GB	15.7 [0.1, 20.6]	246	0.048	0.144	0.432

S8 Full Nested Decision Results

Table S9. Unused life of timely-alerted engines (held-out misses) and fold margins for every lead time and calibration miss target under nested calibration. Engines per subset: 100, 260, 100 and 249. Fixed age: zero-miss fixed-age replacement calibrated on the same folds.

L	Target	Subset	LR	Ridge	RF	GB	CNN	Fixed age
10	0%	FD001	18.4 (1) m27–32	18.9 (1) m27–33	12.2 (0) m14–16	12.6 (0) m11–16	7.4 (4) m15–16	77.8 (1)
		FD002	17.8 (1) m27–31	17.9 (1) m27–31	15.7 (0) m17–20	17.0 (0) m17–20	20.5 (4) m23–27	78.9 (1)
		FD003	25.7 (1) m35–44	25.6 (1) m34–44	12.2 (1) m12–17	12.3 (1) m11–16	21.2 (1) m16–43	102.8 (1)
		FD004	22.6 (0) m32–41	22.6 (0) m32–41	26.6 (2) m16–35	23.2 (1) m19–27	29.4 (0) m27–31	117.9 (1)
	2%	FD001	16.3 (3) m24–28	15.6 (4) m23–28	11.3 (1) m12–14	11.1 (0) m11–14	5.8 (6) m14–15	–
		FD002	15.0 (7) m23–27	15.2 (7) m23–27	12.4 (7) m13–16	13.3 (7) m13–16	16.3 (22) m21–24	–
		FD003	22.9 (2) m32–38	22.3 (2) m31–38	9.9 (1) m10–13	11.2 (2) m10–16	12.5 (1) m14–20	–
		FD004	20.2 (4) m29–34	20.2 (4) m29–34	15.0 (6) m13–18	16.8 (5) m15–18	20.6 (3) m21–25	–
15	0%	FD001	20.2 (1) m31–34	20.8 (1) m31–35	18.9 (1) m19–23	20.8 (1) m19–27	9.2 (3) m13–17	77.8 (1)
		FD002	32.8 (1) m42–49	32.8 (1) m42–49	22.9 (0) m24–31	26.0 (0) m28–32	37.1 (1) m25–44	78.9 (1)
		FD003	32.6 (1) m40–52	32.0 (1) m40–53	19.7 (1) m17–25	18.5 (1) m14–22	27.0 (2) m16–59	102.8 (1)
		FD004	30.3 (1) m40–50	30.3 (1) m40–50	28.9 (1) m27–38	26.6 (2) m21–31	42.1 (1) m24–73	117.9 (1)
	2%	FD001	19.8 (2) m30–33	19.9 (2) m30–33	16.9 (1) m17–21	17.8 (2) m17–20	7.7 (9) m12–16	–
		FD002	18.2 (8) m28–33	18.3 (8) m28–33	18.6 (8) m18–23	18.7 (6) m18–23	19.6 (8) m21–25	–
		FD003	23.2 (1) m32–39	23.1 (1) m32–39	17.2 (2) m15–22	17.3 (2) m14–20	14.9 (2) m16–20	–
		FD004	22.4 (5) m36–37	22.3 (2) m36–37	20.0 (4) m18–22	20.1 (3) m17–21	23.3 (2) m21–25	–
30	0%	FD001	60.9 (1) m49–54	61.3 (1) m49–55	78.3 (0) m64–70	79.1 (1) m61–73	28.5 (2) m19–26	77.8 (1)
		FD002	67.1 (1) m55–60	67.1 (1) m55–60	93.9 (0) m72–76	85.5 (0) m69–74	79.7 (3) m41–76	78.9 (1)
		FD003	42.7 (1) m44–53	43.0 (1) m44–53	50.9 (0) m38–54	45.4 (2) m32–58	34.4 (1) m19–65	102.8 (1)
		FD004	68.2 (1) m53–59	68.2 (1) m53–59	97.9 (1) m70–79	79.4 (0) m67–77	62.9 (0) m38–58	117.9 (1)
	2%	FD001	42.6 (3) m39–45	44.3 (3) m40–46	63.1 (2) m52–65	66.4 (2) m59–67	21.4 (2) m16–22	–
		FD002	50.9 (6) m47–52	51.3 (6) m47–53	64.3 (6) m58–66	57.1 (6) m54–61	41.9 (6) m29–32	–
		FD003	31.5 (3) m36–44	31.9 (3) m36–44	44.1 (1) m35–50	35.8 (2) m31–47	24.3 (2) m17–36	–
		FD004	50.0 (4) m49–54	49.7 (4) m49–54	40.0 (5) m36–48	40.8 (6) m43–47	41.1 (5) m26–30	–

Table S10. Paired differences in unused life over engines alerted in time by both models, with 95% bootstrap intervals including margin re-estimation (* interval excludes zero). Negative values favour the first model. Only the 15-cycle, 0% rows are confirmatory.

L	Target	Subset	RF − GB	GB − LR	RF − LR	CNN − GB	CNN − LR
10	0%	FD001	−0.4 [−2.2, 0.9]	−5.8 [−8.2, −2.7]*	−6.2 [−8.9, −3.1]*	−5.5 [−7.4, −3.7]*	−11.4 [−14.0, −8.1]*
		FD002	−1.2 [−1.9, −0.4]*	−0.8 [−3.1, 0.6]	−2.0 [−4.1, −0.7]*	3.4 [−0.0, 6.2]	2.6 [−1.0, 4.6]
		FD003	−0.1 [−3.1, 1.7]	−13.5 [−16.3, −6.8]*	−13.6 [−17.6, −6.9]*	9.1 [−1.5, 11.2]	−4.2 [−15.3, 2.1]
		FD004	3.3 [−2.9, 4.1]	0.5 [−4.7, 2.7]	3.8 [−7.0, 6.1]	6.2 [2.5, 10.8]*	6.7 [0.4, 9.5]*
	2%	FD001	0.1 [−2.6, 1.1]	−5.4 [−8.8, −2.5]*	−5.3 [−9.7, −2.7]*	−5.7 [−7.5, −3.2]*	−11.2 [−14.0, −7.5]*
		FD002	−0.8 [−2.1, 1.1]	−1.8 [−4.6, 0.8]	−2.6 [−4.8, −0.3]*	2.7 [−0.6, 6.4]	1.1 [−2.5, 4.5]
		FD003	−1.3 [−3.1, 1.9]	−11.8 [−16.9, −6.0]*	−13.1 [−17.8, −6.4]*	1.5 [−2.6, 11.3]	−10.0 [−16.3, 2.8]
		FD004	−1.9 [−3.6, 0.6]	−3.6 [−6.6, 1.2]	−5.4 [−8.1, −0.5]*	3.9 [0.6, 9.5]*	0.4 [−2.6, 7.7]
15	0%	FD001	−1.9 [−4.6, 1.9]	0.6 [−3.8, 2.6]	−1.2 [−4.5, 0.9]	−11.9 [−15.5, −7.5]*	−11.2 [−15.4, −8.5]*
		FD002	−3.1 [−5.0, −1.3]*	−6.7 [−7.8, 1.8]	−9.8 [−11.8, 0.5]	11.0 [−1.4, 13.6]	4.3 [−6.0, 12.5]
		FD003	1.2 [−2.8, 3.3]	−14.1 [−17.6, −2.7]*	−12.9 [−18.0, −1.5]*	8.6 [−5.6, 11.6]	−5.2 [−19.7, 6.4]
		FD004	2.5 [−4.7, 5.1]	−3.8 [−9.3, 3.5]	−1.4 [−8.7, 5.8]	15.7 [0.1, 20.6]*	11.8 [−3.9, 20.2]
	2%	FD001	−0.8 [−5.1, 1.8]	−2.0 [−4.2, 2.8]	−2.7 [−5.2, 1.1]	−11.0 [−15.5, −7.1]*	−12.9 [−15.9, −8.3]*
		FD002	−0.3 [−3.4, 1.9]	0.7 [−3.3, 3.4]	0.4 [−4.9, 2.1]	0.9 [−4.4, 5.8]	1.6 [−5.7, 5.8]
		FD003	−0.1 [−3.7, 3.5]	−6.2 [−18.1, −0.9]*	−6.3 [−18.9, −0.7]*	−2.4 [−6.2, 12.2]	−8.2 [−20.2, 7.8]
		FD004	−0.2 [−4.0, 2.7]	−2.3 [−5.8, 2.4]	−2.5 [−6.0, 2.1]	3.2 [−2.0, 8.1]	0.9 [−3.9, 6.1]
30	0%	FD001	−0.2 [−11.4, 9.7]	18.4 [9.6, 32.1]*	18.1 [12.6, 26.6]*	−52.0 [−62.2, −28.5]*	−33.0 [−40.5, −9.3]*
		FD002	8.4 [−2.1, 14.6]	18.8 [6.4, 22.8]*	27.2 [12.7, 30.1]*	−6.7 [−23.8, 6.5]	12.2 [−6.6, 20.4]
		FD003	6.4 [−2.3, 17.0]	2.3 [−1.8, 8.6]	8.6 [−1.2, 24.4]	−10.5 [−23.6, 2.6]	−7.8 [−20.7, 7.5]
		FD004	18.2 [4.0, 22.0]*	11.6 [−26.0, 21.4]	29.8 [−18.4, 40.7]	−16.5 [−27.0, 18.3]	−5.1 [−19.0, 6.4]
	2%	FD001	−3.4 [−10.9, 10.6]	24.4 [5.6, 31.8]*	21.4 [12.3, 30.5]*	−45.6 [−60.5, −18.0]*	−21.1 [−38.7, −4.6]*
		FD002	7.3 [−1.6, 10.0]	6.2 [−1.9, 20.7]	13.5 [1.0, 23.7]*	−15.5 [−31.8, 3.7]	−9.0 [−20.7, 12.1]
		FD003	8.8 [−3.3, 16.7]	4.6 [−1.4, 8.8]	13.5 [−2.1, 23.8]	−11.4 [−24.5, 2.9]	−6.6 [−21.6, 7.8]
		FD004	−0.6 [−2.2, 17.4]	−9.6 [−21.0, 11.2]	−10.3 [−18.6, 26.7]	0.5 [−21.7, 15.7]	−9.0 [−21.8, 6.0]

Code, model predictions and all intermediate results: https://github.com/toochii2/isdia2027-rul-decision-evaluation (archived at https://doi.org/10.5281/zenodo.23378792)